

Disproof of Conjecture 00000005244

earthking11

October 6, 2026

The two filed clauses

Write $c(g)$ for the number of irreducible components at genus g . Under the terminology of the file, irreducibility means $c(g) = 1$.

The first clause says that the moduli space is always irreducible for genus at least two:

$$\forall g \geq 2, \quad c(g) = 1.$$

The next clause says that genus two is exceptional and decomposes into two irreducible components:

$$c(2) = 2.$$

Proposition 1. *No component-count function can satisfy both clauses.*

Proof. Because $2 \geq 2$, substituting $g = 2$ into the universal first clause gives $c(2) = 1$. The second clause gives $c(2) = 2$. Thus $1 = 2$, a contradiction. $\square$

The later assertions about the component dimensions, their intersection, and the discriminant degree do not affect this contradiction. This disproof is about the conjunction as filed; it does not choose which of the two mutually incompatible intended statements should replace it. A syntactic repair would at least have to change “genus at least 2” to a range excluding genus 2, or remove the claimed genus-two decomposition.

The accompanying Lean theorem quantifies over every possible component-count function and proves that the conjunction is impossible.